
Input: Code parameters $(n, k=1)$; HSE depth d ; number of seeds M ; fault set $\mathcal{F}$ (all $3n$ weight-one Paulis);
adversary mode $\in \{\text{ADAPTIVE}, \text{COMMITTED}\}$; validated DRBG

Output: Mean accepted logical disturbance $\bar{\Delta}$; fraction ϕ_{KL} of encoders satisfying Equation (7.2)

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1  $\Sigma \leftarrow 0$ ;  $\Phi \leftarrow 0$ ;  
2 if  $mode = \text{COMMITTED}$  then adversary commits to  $P^* \in \mathcal{F}$  before any seed is drawn;  
3 for  $m \leftarrow 1$  to  $M$  do  
4    $s_m \leftarrow \text{DRBG.GENERATE}()$ ; // SP 800-90A instance inside a FIPS 140-3 module  
5    $E_m \leftarrow \text{HSE}(s_m, d)$ ; derive  $\mathcal{S}_m, \bar{X}_m, \bar{Z}_m$  by tableau conjugation; // Equations (7.3) and (7.8)  
6   encode  $|\psi\rangle |0^{n-1}\rangle \mapsto E_m(|\psi\rangle |0^{n-1}\rangle)$  on the QPU;  
7   if  $mode = \text{ADAPTIVE}$  then  
8     | adversary learns  $E_m$  and injects  $P_m \leftarrow \arg \max_{P \in \mathcal{F}} \Delta(E_m, P)$ ;  
9   else  
10    | adversary injects  $P_m \leftarrow P^*$ ;  
11   measure the syndrome  $\sigma(P_m)$  against  $\mathcal{S}_m$ ; accept iff  $\sigma = 0$ , otherwise reject;  
12    $\Sigma \leftarrow \Sigma + \Delta(E_m, P_m)$ ; // Equation (7.4); zero on rejection  
13    $\Phi \leftarrow \Phi + \mathbb{I}[E_m \text{ satisfies Equation (7.2) for } \mathcal{F}]$ ;  
14   append  $(m, \text{id}(s_m), \sigma(P_m), \Delta(E_m, P_m))$  to the signed audit log;  
15 return  $\bar{\Delta} \leftarrow \Sigma/M, \phi_{\text{KL}} \leftarrow \Phi/M$ ;
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